

ADBMS MINI PROJECT REPORT

Finance Investment Survey Analysis

Project Details

Name : Dhanashri Rajesh Malvadkar
Subject: Advanced Database management system
Dataset: Finance Investment Survey (40 Respondents)
Tools: Microsoft Excel | PostgreSQL | Power BI
Year: 2025-26

1. Problem Statement & Motivation

In today’s world, many young people are starting to invest their money, but not everyone understands how they make their decisions. People between the age of 21 to 35 are becoming active investors, but their choices and thinking patterns are not clearly studied.

So, in this project, I tried to analyze a survey dataset of 40 people to understand:

- Which investment options are most preferred
- How age and gender affect investment decisions
- What people expect from their investments
- Where they get their financial information from
- How long they prefer to invest

This analysis can help in understanding how young investors think and what they actually want while investing.

2. Dataset Description

The dataset used in this project is a primary survey dataset collected from 40 young investors. It contains 24 columns covering demographics, investment preferences, behavioural patterns, and reasoning behind investment choices.

Attribute	Details
Total Records	40 investors
Total Columns	24 columns
Age Range	21 to 35 years
Gender	Male (25), Female (15)
Data Source	Primary survey data (Kaggle)
File Format	CSV (Finance_data.csv)
Key Columns	gender, age, preferred_avenue, objective, purpose, duration, expect, factor, info_source
Rating Columns	Mutual_Funds, Equity_Market, Debentures, Government_Bonds, Fixed_Deposits, PPF, Gold (ranked 1-7)

3. Tools & Technologies Used

This project follows a real-world data pipeline using three industry-standard tools:

Tool	Purpose	What We Used It For
Microsoft Excel	Data Collection & Cleaning	Removed duplicates, fixed data types, added Age_Group column, created pivot tables and EDA summary
PostgreSQL 18	Database Storage & SQL Analysis	Designed 2 normalized tables, wrote 22+ SQL queries, created 3 views, applied indexes and window functions
Microsoft Power BI	Data Visualization & Dashboard	Connected to PostgreSQL, built 5 interactive visuals with 2 slicers for dynamic filtering

4. Data Collection Process

The dataset was sourced from Kaggle — a public data platform widely used for academic and research projects. It represents a real-world survey conducted among young investors regarding their investment habits and preferences.

4.1 Dataset Source

- Platform: Kaggle (www.kaggle.com)
- File name: Finance_data.csv
- Size: 40 rows x 24 columns
- Type: Structured tabular data (survey responses)

4.2 Column Categories

- Demographics: gender, age
- Investment preferences: preferred_avenue, objective, purpose, duration
- Behavioural: factor, invest_monitor, expect, stock_market
- Ratings: mutual_funds, equity_market, debentures, govt_bonds, fixed_deposits, ppf, gold (ranked 1-7, lower = more preferred)
- Reasoning: reason_equity, reason_mutual, reason_bonds, reason_fd
- Other: info_source, savings_objective

5. Data Cleaning Steps

Before using the data, I cleaned it in Excel to make sure there are no errors.

Some steps I followed:

Checked for missing values → No missing values found

Checked for duplicate rows → No duplicates

Verified data types → All numeric values were correct

Found a typo in “Stock_Markt” → kept it same but handled later

Renamed long column names for easier use in SQL

Created a new column called Age_Group (21–25, 26–30, 31–35)

Even though the dataset looked clean, I still checked everything carefully.

6. Database Design & SQL Logic

6.1 Database Design

The cleaned data was stored in PostgreSQL using a normalized 2-table design:

- Table 1: investors — stores all demographic and behavioural data (19 columns)
- Table 2: investment_ratings — stores the 7 investment ratings per investor (9 columns)
- Relationship: investment_ratings.investor_id references investors.investor_id (Foreign Key)
- The age_group column is auto-generated using GENERATED ALWAYS AS — no manual entry needed

6.2 Key SQL Queries Written

#	Query Type	Purpose
1	Basic Aggregations	COUNT, AVG, GROUP BY on gender, avenue, objective, duration, expect
2	JOIN Queries	Combined investors + investment_ratings to get full investor profiles
3	Filter Queries	Filtered by age group, gender, expected returns, information source
4	Views	Created 3 reusable views: investor_summary, avenue_gender_summary, agegroup_behavior
5	Window Functions	RANK(), COUNT() OVER, AVG() OVER, SUM() OVER for advanced analysis
6	Indexes	Created 5 indexes to improve query performance on frequently filtered columns

6.3 Sample SQL Query

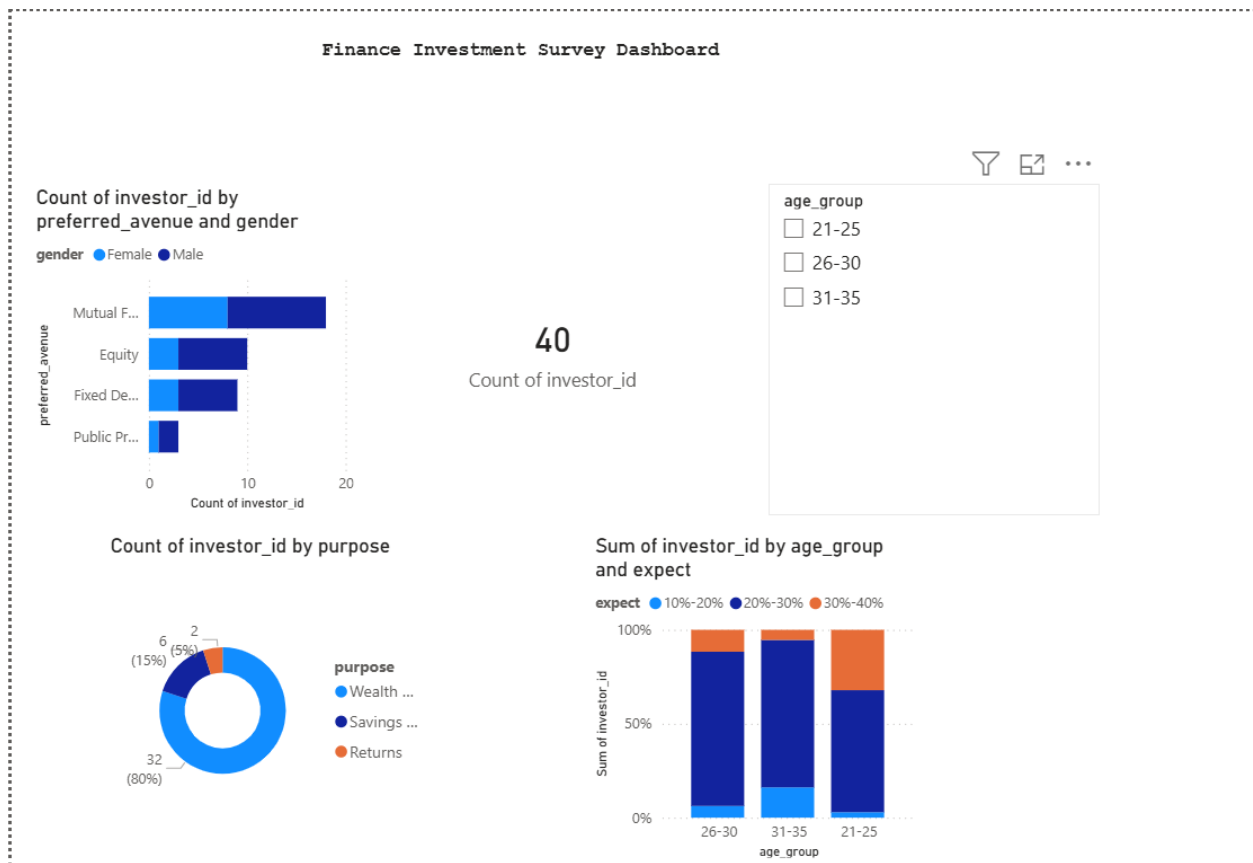
The following query was used to generate the data for Power BI:

```
SELECT i.investor_id, i.gender, i.age, i.age_group, i.preferred_avenue, i.objective, i.expect,
r.mutual_funds, r.gold FROM investors i JOIN investment_ratings r ON i.investor_id = r.investor_id
ORDER BY i.investor_id;
```

7. Power BI Dashboard & Insights

7.1 Dashboard Overview

The Power BI dashboard was built by connecting directly to the PostgreSQL database using the investor_summary view. The dashboard contains 5 interactive visuals and 2 slicers for dynamic filtering.



7.2 Visuals Built

#	Visual Type	Title	Fields Used
1	Clustered Bar Chart	Avenue by Gender	preferred_avenue, investor_id (Count), gender
2	Donut Chart	Investment Purpose	purpose, investor_id (Count)
3	Card (KPI)	Total Investors	investor_id (Count) = 40
4	Stacked Column	Returns by Age Group	age_group, investor_id, expect
5	Slicer	Age Group Filter	age_group

8. Key Insights from the Analysis

After doing analysis in Excel, SQL, and Power BI, I found some interesting points:

- Most respondents were male (around 62%)
- Average age is around 27–28 years
- Mutual funds are the most preferred investment
- Most people invest for wealth creation
- Many expect high returns (20–30%)
- Majority prefer investing for 1–5 years
- Financial consultants are a major source of information
- Gold is still a popular traditional investment
- Government schemes like PPF are less preferred

These insights show that young investors are more focused on growth and returns rather than safety.

9. Learnings & Challenges Faced

- What I learned:

How to clean and prepare real data
Writing SQL queries like JOIN, GROUP BY, etc.
How to connect Power BI with database
How to convert data into meaningful insights

- Challenges faced:

Faced error while importing CSV into PostgreSQL
Had issue with foreign key while inserting data
Power BI was showing wrong aggregation (Sum instead of Count)

But I solved these by checking errors and understanding concepts properly.

10. Conclusion & Future Improvements

10.1 Conclusion

In this project, I learned the complete process of data analysis starting from raw data to final dashboard.

I used:

Excel for cleaning

PostgreSQL for analysis

Power BI for visualization

Overall, this project helped me understand how data is used in real-world situations and improved my practical skills.

10.2 Future Improvements

- Collect a larger dataset (500+ respondents) for more statistically reliable insights
- Add time-series data to track how investment preferences change over the years
- Include income level data to understand how earnings affect investment choices
- Build predictive models using Python/ML to predict preferred avenue based on demographics
- Automate the data pipeline using Python scripts to refresh Power BI daily
- Expand geographic coverage to compare urban vs rural investor preferences